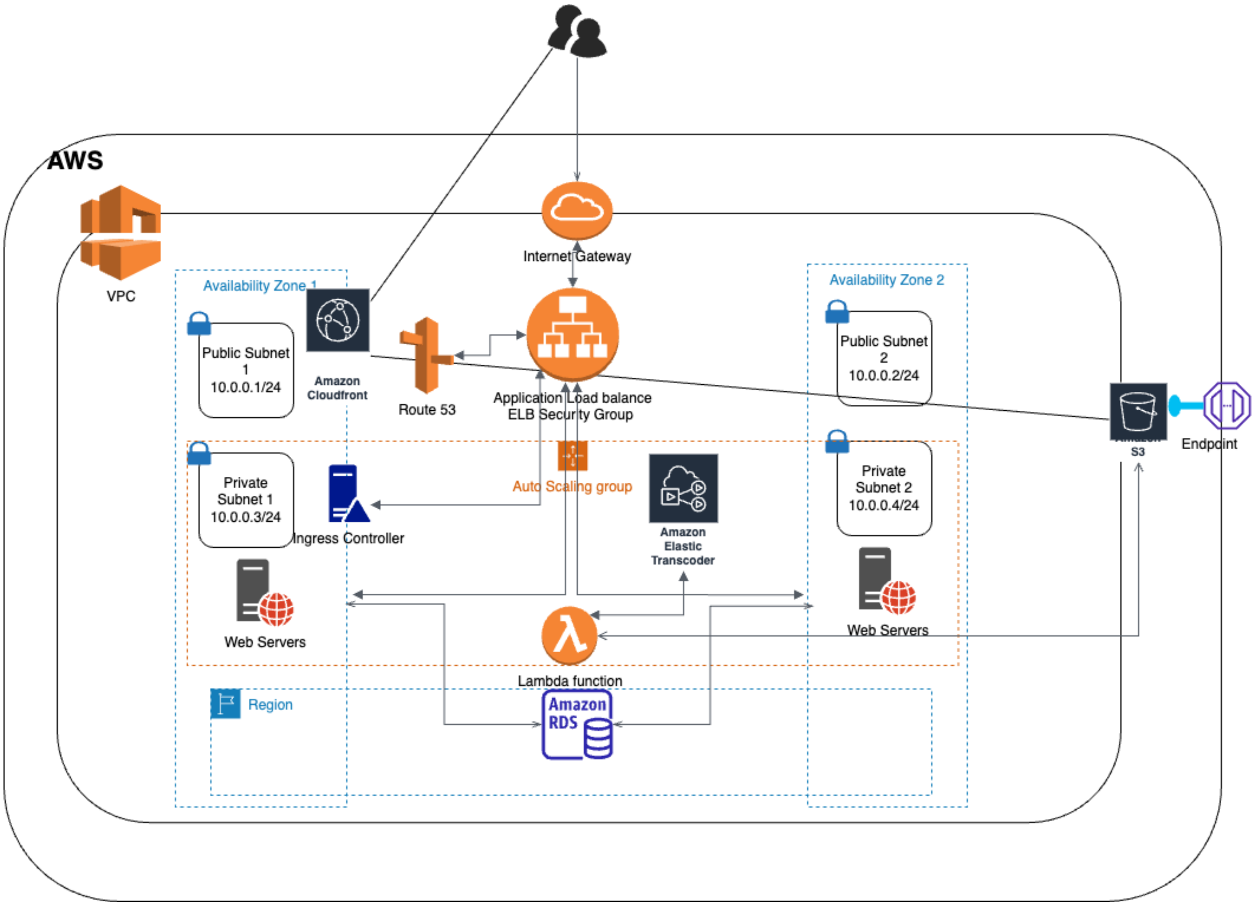
Architecture:



Design Rationale

AWS S3: Although containers offer flexibility, they are not the ideal solution for our situation. For our application to run on a managed cloud service without requiring us to manage the underlying infrastructure, we specifically need a serverless/event-driven solution. With containers, it is not possible because managing the host machines is still necessary. So, for the solution, Amazon S3 can be used which is highly durable, scalable, and cost-effective. Additionally, it offers universal access to media files, which is necessary for users in various nations to access the application. AWS Lambda and other AWS services are integrated with S3.

AWS Lambda: It is highly scalable, and automatically scales up or down based on the demand. Lambda functions can be triggered by various events, including file uploads to S3, which makes it perfect for processing media files in real time. Lambda functions can be written in various programming languages, and developers only pay for the compute time used by each function, making it a cost-effective solution for the Photo Album application.

Amazon DynamoDB: In our case, we have a relatively simple table structure and want to explore more cost-effective options. NoSQL databases, like Amazon DynamoDB, maybe a better fit for our scenario as they are more scalable, performant, and cost-effective for simple data models. DynamoDB is a NoSQL database that is highly scalable and performs well at any scale. It is also fully managed, meaning that the company does not have to worry about database administration. DynamoDB can be used to store metadata about photos and other media files and can be easily integrated with other AWS services.

Security Groups: In the photo album application, we can create separate security groups for the different types of instances, such as the web server, application server, and database server. The security groups should allow only the necessary traffic to enter and exit the instances, such as HTTP, HTTPS, SSH, and database traffic. Using least privilege principles to limit user access to the application and its resources. ACLs can be used in conjunction with security groups to provide an additional layer of security.

Amazon CloudFront: When a user requests content from an application, CloudFront can cache the requested content at the edge location that is most convenient for the user, speeding up the user's access to the requested content. To guarantee that users receive content securely, CloudFront also supports the use of HTTPS for secure content delivery. CloudFront can be configured to use the appropriate caching behavior, TTL, and invalidation settings for each type of content, ensuring that content is delivered efficiently and in a timely manner. By using CloudFront, the application can improve global response times, reduce latency, and ensure that content is delivered securely to users.

AWS Elastic Transcoder: In the scenario described, the company wants various versions of media to be automatically produced when a media item is uploaded to the S3 bucket. Elastic Transcoder can be used to automatically transcode media files into multiple formats and resolutions as required. For example, when a user uploads a high-resolution video file to the S3 bucket, Elastic Transcoder can automatically transcode it into a lower-resolution version that is suitable for mobile devices. It can also create multiple output versions of the same file, such as thumbnails and alternate formats.

One advantage of using Elastic Transcoder is that it is a fully managed service, which means that the company does not need to worry about managing the underlying infrastructure or scaling the service to handle large volumes of media files. Elastic Transcoder also integrates seamlessly with other AWS services, such as S3 and CloudFront, making it easy to incorporate into the existing architecture.

Amazon SQS: For the photo album website SQS can be used for decoupling the processing of media transformation jobs. The production of alternate versions should be initiated immediately when a media item is uploaded to the S3 bucket, and these modified media will also be kept in S3. You can construct a queue for media transformation jobs using Amazon SQS. For each piece of media that is uploaded to S3, the application can push messages to the queue, and worker nodes can process these messages by pulling them from the queue. By doing this, the application is effectively detached and is prevented from becoming overloaded. The queued transformation jobs can be processed by several worker nodes, each of which is capable of specialising in different tasks.

Performance, scalability, and reliability: The use of serverless computing, caching, and queueing can improve the performance, scalability, and reliability of the application. Lambda functions can scale automatically based on demand, while CloudFront can cache frequently accessed files and reduce the load on the backend. SQS can decouple components and improve reliability by providing fault tolerance.

Route 53:

Route 53 can be used to manage DNS and domain name registration for the Photo Album application. It can route traffic to the application, balancing it across multiple EC2 instances or Kubernetes clusters in different regions to ensure high availability.

Cognito:

Cognito can be used to authenticate and authorize users of the Photo Album application. It can provide features such as user sign-up, sign-in, and access control, as well as integration with social identity providers, such as Facebook and Google.

Security: AWS provides various security features, including encryption, access control, and monitoring the following security measures should be taken:

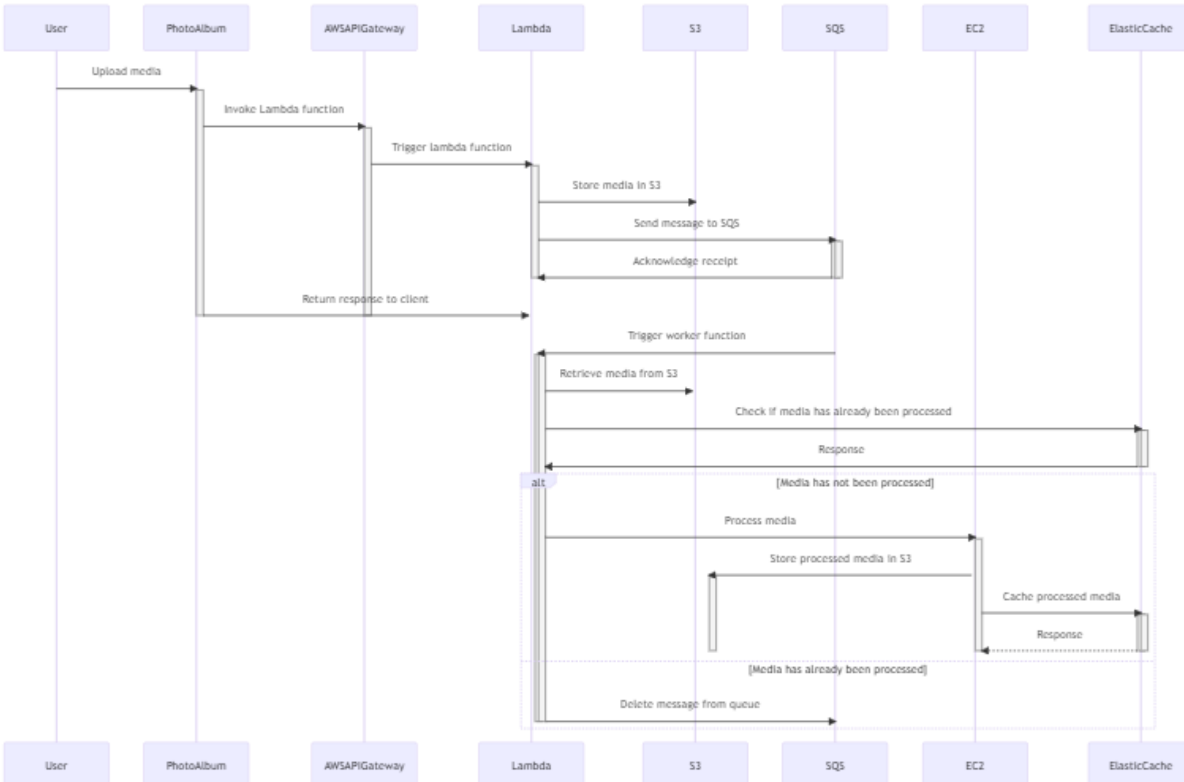
- VPC Security Groups and Network ACLs: As mentioned earlier, the VPC security group and network ACLs should be configured to restrict access to only authorized IP addresses, protocols, and ports. This ensures that only authorized users and services can access the application.

- Encryption: All data should be encrypted at rest and in transit. AWS provides various encryption options, such as server-side encryption for S3 buckets, SSL/TLS for CloudFront and API Gateway, and AWS Key Management Service (KMS) for managing encryption keys.

- Access Control: Access to the application and its resources should be controlled using IAM (Identity and Access Management) policies. IAM policies should be used to grant only the necessary permissions to users, roles, and services.

- Logging and Monitoring: Logging and monitoring should be set up to detect and respond to security incidents. AWS provides various logging and monitoring services, such as AWS CloudTrail, Amazon CloudWatch, and AWS Config.

Sequence Diagram



The sequence diagram depicts the interactions between the system components (S3 Bucket, Lambda, Media Processing Service) and the actors (User and Photo Album Application) involved in the media processing workflow. The workflow is started by the user by uploading media (such as a photo) to the S3 Bucket. The S3 Bucket receives a request to upload the files from the Photo Album Application. When media is uploaded to the S3 Bucket, a Lambda function is called, which initiates the media processing workflow. To process the media, the Lambda function sends a message to the Media Processing Service after extracting the metadata from the media file. The media file after processing is saved to S3 the worker node saves the processed file to the S3 Bucket after processing the media file. The Lambda function notifies the Photo Album Application that the media file has been processed and is now ready for viewing. The user can now see the processed media in the Photo Album Application. On the Photo Album application, the user can now view the processed media file (such as a thumbnail).